

Haplic Yermosols

Description:

Haplic Yermosols are typical desert soils formed under hyper-arid to arid climatic conditions where annual rainfall is usually less than 250–500 mm.

The term “*Yermosol*” comes from the Greek *yermos*, meaning “barren,” indicating soils with very low biological activity and weak horizon development.

“Haplic” designates typical representatives of this group without strong salic, calcic, or gypsic accumulations. They form mainly from unconsolidated alluvial, colluvial, or aeolian materials on plains, terraces, and foothills. The surface layer is often sandy or loamy, pale brown to light grey, and structureless, with low organic matter and high mineral content.

Drainage:

- **Wet season:** Generally well to excessively drained due to coarse texture and low rainfall. Water infiltration is rapid, but storage is minimal.
- **Dry season:** Very dry for most of the year; moisture stress is the main limiting factor for vegetation and crop growth.

Landscape:

Widespread in arid and semi-arid landscapes, including desert plains, piedmont slopes, and alluvial fans. In Kenya, Haplic Yermosols are common in northern and northeastern regions such as Turkana, Marsabit, Wajir, and Garissa counties. They occur on gently sloping to flat terrain with sparse vegetation cover, mainly shrubs and grasses adapted to drought.

Depth:

Shallow to moderately deep (20–80 cm), depending on erosion and parent material. Rooting depth may be restricted by compacted subsoil, calcareous hardpan, or stoniness.

Workability:

- **When moist:** Easy to cultivate due to loose texture and low cohesion; however, moisture availability is limited.
- **When dry:** Hard and crusted; surfaces may seal, hindering infiltration and seedling emergence. Wind erosion risk is high if soils are left bare.

Nutrient Richness & Cation Exchange Capacity (CEC):

- **CEC:** Very low to low (3–10 cmol(+)/kg), reflecting the dominance of quartz and minimal clay or organic matter.
- **Base saturation:** High (>80%) because of calcium and magnesium salts retained by minimal leaching.
- **Nutrients:** Nitrogen and phosphorus are extremely deficient; potassium is moderate but poorly available. Micronutrient deficiencies (Fe, Zn) can occur due to alkalinity.
- **Organic matter:** Extremely low (<1%), resulting in weak structure and limited biological activity.

pH:

Neutral to moderately alkaline (7.0–8.5), with occasional localized salinity or carbonate accumulations.

Management:

- Incorporate organic amendments (compost, manure, crop residues) to improve water retention and fertility.
- Employ mulching and minimum tillage to reduce evaporation and surface crusting.
- Use drip irrigation or water-harvesting techniques (trenches, bunds, zai pits) for moisture conservation.
- Apply nitrogen and phosphorus fertilizers in small, frequent doses to improve uptake efficiency.
- Introduce drought- and heat-tolerant crop varieties.
- Establish windbreaks and maintain vegetative cover to control wind erosion.
- Avoid overgrazing to protect surface structure and prevent desertification.

Suitable Crops:

Best suited for hardy, drought-tolerant crops such as sorghum, millet, cowpeas, green grams, and forage grasses. With proper irrigation management, horticultural crops like onions, tomatoes, and melons can also be grown successfully.